

Artificial Intelligence & Machine Learning – Task 3 Report

Model Validation, Overfitting Control & Hyperparameter Tuning

1. Introduction

Machine learning models should not only perform well on training data but also make accurate predictions on new, unseen data. In this task, model validation techniques were used to evaluate the model's performance and reduce overfitting. Cross-validation and hyperparameter tuning were also applied to improve the model's reliability.

2. Objective

The objective of this task was to detect overfitting, validate the model using Cross Validation, improve its performance through Hyperparameter Tuning, and compare different regression models to select the most suitable one.

3. Dataset Used

The **California Housing Dataset** from Scikit-learn was used for this project. The dataset contains housing-related features such as median income, house age, average rooms, population, latitude, and longitude. The target variable is the house price.

4. Implementation Summary

The dataset was loaded and the features were scaled using **StandardScaler**. After splitting the dataset into training and testing sets, a **Decision Tree Regressor** was trained. The model was evaluated using RMSE and R^2 Score. Cross Validation was then performed to obtain a more reliable estimate of the model's performance. Finally, **GridSearchCV** was used to find the best values for **max_depth** and **min_samples_split**. The optimized Decision Tree model was then compared with Linear Regression and Ridge Regression.

5. Overfitting Analysis

The initial Decision Tree model showed signs of overfitting. The Training RMSE was almost **0**, while the Testing RMSE was **0.702173**, indicating that the model had learned the training data too closely and did not generalize well to unseen data.

6. Cross Validation Results

Five-fold Cross Validation was performed to evaluate the model more reliably.

Cross Validation RMSE Scores:

- 0.877525
- 0.817541
- 0.905193
- 0.955814
- 0.924714

Average Cross Validation RMSE: 0.896157

These results show that the model performs consistently across different data splits.

7. Hyperparameter Tuning

GridSearchCV was used to optimize the Decision Tree model.

Best Parameters:

- **max_depth:** 10
- **min_samples_split:** 10

After tuning, the optimized model achieved:

- **RMSE:** 0.645430
- **R² Score:** 0.682099

This shows an improvement over the initial Decision Tree model.

8. Model Comparison

Model	RMSE	R ² Score
Linear Regression	0.745581	0.575788
Ridge Regression	0.745554	0.575819
Decision Tree (Tuned)	0.645430	0.682099

The Tuned Decision Tree performed better than the other two models by achieving the lowest RMSE and the highest R² Score.

9. Final Model Selection Justification

The Tuned Decision Tree Regressor was selected because it provided the best overall performance among the three models. Hyperparameter tuning reduced overfitting, and Cross Validation helped confirm that the model performs consistently on different subsets of the data. Although Linear Regression and Ridge Regression are simpler models, the Tuned Decision Tree achieved better prediction accuracy, making it the most suitable choice for this task.

10. Conclusion

In this task, model validation techniques were successfully applied to improve the performance of a Decision Tree Regressor. Cross Validation and Hyperparameter Tuning helped reduce overfitting and improve the model's reliability. After comparing different regression models, the Tuned Decision Tree was selected as the best-performing model based on its RMSE and R² Score.